Algebra SEP Worksheet 9 8/15/23

Rings and Modules

Pride

1. Find an example of each of the following.

(a) **(Aug 2018 1(a))** A short exact sequence of modules that is not split exact.

(b) A projective module that is not free.

(c) A flat module that is not projective.

(d) **(Aug 2018 1(b)) and (Aug 2020 1(d))** A flat $\mathbb{Z}$-module that is not free.

(e) Nonzero modules M, N over a ring R such that $M \otimes_R N = 0$.

(f) A ring R and an R-algebra A that is finitely generated as an algebra over R but is not finitely generated as an R-module.

2. (August 2017 Problem 1) Let $M = \mathbb{C}[x]/(x^2 + x)$ and $N = \mathbb{C}[x]/(x - 1)$. Compute the dimension of the following tensor products as vector spaces over $\mathbb{C}$:

(a) $M \otimes_{\mathbb{C}[x]} N$

(b) $M \otimes_{\mathbb{C}} N$

(c) $M \otimes_{\mathbb{C}[x^2]} N$.

3. (January 2019 Problem 1)

(a) Give an example of a simple ring which is not a commutative ring.

(b) Give an example a ring R, R-modules M_1, M_2, M_3, and N and a short exact sequence $0 \to M_1 \to M_2 \to M_3 \to 0$ such that the corresponding sequence $0 \leftarrow \mathrm{Hom}(M_1, N) \leftarrow \mathrm{Hom}(M_2, N) \leftarrow \mathrm{Hom}(M_3, N) \leftarrow 0$ is not exact.

The Fall

4. (August 2020 Problem 4) Let R be a commutative ring.

(a) Give an example of a free R-module M and an injective homomorphism $M \to M$ which is not an isomorphism. (For this part, R can be whatever commutative ring you like.)

(b) If M_1 and M_2 are free of finite ranks r_1 and r_2, show that a surjective map $M_1 \to M_2$ exists if and only if $r_1 \geq r_2$.

(c) Assume furthermore that R is an integral domain. Show that an injective map $M_1 \to M_2$ exists if and only if $r_1 \leq r_2$.

5. (January 2022 Problem 3) Let R be a commutative ring.

(a) Prove that a module M over R is projective if and only if it is a direct summand of a free module.

(b) Let M be a projective module over R and $S \subset R$ a multiplicative set. Prove that $S^{-1}M$ is a projective $R[S^{-1}]$-module.

6. Show that if R is a PID (principal ideal domain) and M is an R-module, then M is flat if and only if M is torsion free.

7. (January 2021 Problem 3)

(a) Show that a projective module over an integral domain is torsion free.

(b) Let $0 \to M_1 \to M_2 \to M_3 \to 0$ be a short exact sequence of finitely generated modules over a commutative, Noetherian ring. Prove the following OR give a counterexample:

  (i) If M_1 and M_2 are projective, then so is M_3.

  (ii) If M_2 and M_3 are projective, then so is M_1.

8. (August 2021 Problem 4) Decide if the following statements are true. If so, provide a proof, if not, provide a counterexample. Let M, N be finitely generated modules over a commutative ring A. Let $f\colon M \to N$ be a homomorphism of A-modules, and let $g\colon B \to A$ be a homomorphism of commutative rings.

(a) "If f is injective, so is the induced morphism $M \otimes_A M \to N \otimes_A N$."

(b) "If f is surjective, so is the induced morphism $M \otimes_A M \to N \otimes_A N$."

(c) "If g is injective, so is the induced morphism $M \otimes_B M \to M \otimes_A M$."

9. (August 2017 Problem 5) (This problem has been slightly modified from its original form.) Let $B = \mathbb{Q}[x, y]/(xy)$.

(a) Consider the map $\mathbb{Q}[t] \to B$ sending t to x. This map makes B into a $\mathbb{Q}[t]$-module. Show that B is *not* a finite $\mathbb{Q}[t]$-module under this map.

(b) Consider the map $\mathbb{Q}[t] \to B$ sending t to $x + y$. Show that this map makes B into a finite $\mathbb{Q}[t]$-module.

(c) Consider B as a $\mathbb{Q}[t]$-module under the map from part (a). Is B a flat $\mathbb{Q}[t]$-module? What about when considered as a $\mathbb{Q}[t]$-module under the map from part (b).